Binghuan (Warner) Wu

warner0709@outlook.com
warner-ng.github.io
Warner-Ng

And most importantly, have the courage to follow your heart and intuition.

Education

2026 **Visiting Student**, *University of California, Berkeley*
Courses: Nonlinear Control (MECENG C237), Deep Reinforcement Learning (COMPSCI 185/285)

2023–Present **B.E. in Control Engineering**, *Tongji University*, China
GPA 4.4/5.0

Publications

Conference

[C3] **Binghuan Wu**, Yiren Rong, Junyang Wu, and Koushil Sreenath.
AlignMimic: Learning Visual Alignment for Precise Humanoid-Object Interaction. *Under review.*

[C2] Yiren Rong, Tse-Wen Chen, Matei Gardea, **Binghuan Wu**, and Koushil Sreenath.
GenPlan: Generative Scene-Grounded Motion Planning and Execution for Humanoid Robots. *Under review.*

[C1] Kaiyang Ji*, Bingsheng Qian*, **Binghuan Wu**, Kangyi Chen, Ye Shi, and Jingya Wang.
DiscoForcing: A Unified Framework for Real-Time Audio-Driven Character Control with Diffusion Forcing. *ICML 2026*

Research Experience

2026 **Research Assistant**, *Hybrid Robotics Lab, University of California, Berkeley*
Advised by: Prof. Koushil Sreenath

- Developed AlignMimic, a perceptive dual-policy framework for long-horizon, fine-grained humanoid-object interaction under object misalignment and external disturbances.
- Designed a task-specific whole-body interaction policy, a task-agnostic recovery policy, and a hysteretic selector to smoothly coordinate manipulation and locomotion alignment.
- Distilled privileged control and switching signals into egocentric depth-based visuomotor policies for deployable closed-loop execution on tasks including bike racking, chair relocation, and box transportation.

2025–2026 **Research Assistant**, *ShanghaiTech University*
Advised by: Prof. Ye Shi and Prof. Jingya Wang

- Conducted research on real-time generation of humanoid motion conditioned on dynamic environmental inputs.
- Trained generative models and designed a model architecture for streaming generation under disruptions.
- Developed a structural music encoder to provide strong supervision for motion generation.
- Implemented a ROS 2 framework for motion retargeting and tracking.

2024–2025 **Research Intern**, *Shanghai Artificial Intelligence Laboratory*
Advised by: Dr. Jingbo Wang and Dr. Jiangmiao Pang

- Independently explored the use of a differentiable physics simulator for robot training.
- Trained a Unitree H1-2 humanoid to kneel and walk.
- Applied reinforcement learning-based motion retargeting to enable the robot to imitate motion priors from AMASS and HUMOTO datasets.
- Contributed engineering code for object-centric motion retargeting for human-object interaction research.
- Applied optimization-based robot control methods to motion retargeting.

2024 **Research Assistant**, *Institute of Precision Optical Engineering, Tongji University*
Advised by: Prof. Xiao Deng

- Acquired fundamental research skills, including paper reading and summarization, and mathematical modeling using MATLAB.
- Contributed to the mathematical modeling of ECDL lasers by developing MATLAB code for simulation and analysis.

2023–2024 **Project Leader**, *Autonomous Driving Group, Tongji University Lotus Racing Team*

- Contributed to a Hough-transform-based algorithm for path planning, improving performance by replacing the original MSE-based method.
- Implemented motion planning and control strategies, including the Stanley method, pure pursuit, and incremental PID.
- Performed on-vehicle testing and debugging and collected LiDAR point cloud data. Evaluated the point cloud clustering algorithm using ROS RViz.

Technical Skills

Programming	C++, Python, Swift, MATLAB, Verilog
Fabrication	Electrical Design
Tools	Linux, ROS, CUDA (Basic), PyTorch, Isaac Sim/Gym
Control	PID, Stanley method, Kalman filter (for an autonomous race car), MPC (Basic)

Projects and Awards

[1] Reproducing the PULSAR drone system from the University of Hong Kong.

[2] Built a low-cost quadrupedal robot equipped with a robotic arm and trained a whole-body control policy.

[3] Designed an LED chaser circuit in the shape of the OpenAI logo.

[4] Built an iOS application centered on Tongji University.

[5] 3rd Prize in Applied Mechanics, Tongji University, 2024.

[6] Merit Award in Structural Mechanics, Tongji University, 2024.

Student Services

2023–Present **Student Staff**, Students' Association for Science and Technology of Tongji Univ.

2023–Present **Long-term Volunteer**, Automobile Culture Promotion Society of Tongji Univ.